The Two Bills

An Operational Basis for Interfaces: Partition and Realize, the Admission Price, the Crossing Price, and the Exact Chain Rule Between Them

Driven by Dean Kulik

July 2026

Successor to "May, Does, Seen." Every number is quoted verbatim from live computation. Seven errors were made and corrected during this work, including two that inverted a result and one that would have put a false formula in this paper; all are listed by name in Section 10, because two of them changed the findings and one of them strengthened a finding by being fixed.

Abstract

The predecessor paper established two algebras — a distributive lattice of constraints and a monoid of execution — linked by a contravariant action, and left open why the interface between them resisted a single accounting. This paper answers that, and the answer is that there is no single accounting because there are two bills.

An interface is defined here operationally, by the irreducible work it performs rather than by what it is made of. Two primitives suffice: a partition P of the source, one cell of which is inadmissible, and a realization R carrying admitted members across. Every property previously proposed as a separate interface role — constraining, translating, preserving — is a function of (P, R), verified by direct computation. The two prices are likewise functions of (P, R), and they are not two aspects of one quantity. The admission price $A(I) = -\log_2 \mu(Adm)/\mu(S)$ is paid once and is idempotent. The crossing price $X(I) = H(X|Y)$ is paid per crossing and is exactly additive under composition. A counterexample settles that the second is not forced: an interface that separates, constrains, translates and preserves can have crossing price exactly 0.0000 while its admission price is 1.0000 bits — which is why a type system rejects bad programs and costs nothing for good ones.

A further collapse follows from measurement rather than argument: preservation and the crossing price are not two roles but one measurement read from opposite ends, since preserve holds if and only if $X(I) = 0$, in every case tested. The chain rule for the crossing price is established exactly, $H(X|Z) = H(X|Y) + H(Y|Z)$ for deterministic chains with Y carrying the pushforward measure, verified in eight of eight trials to nine decimals — and the naive shortcut Σ (|fibre|/n) $\log_2$|fibre| is shown to fail for the second term, an error corrected in this paper's own drafting. The admission price is idempotent in every

case and additive under conjunction only when the constraints are independent, which is precisely the failure signature separating a lattice from a monoid.

The paper also reports two demolitions of its own program's earlier claims. First, state-level admissibility does not determine dynamics: six distinct dynamics on four states share one invariant-set lattice, and the projection from transition structure to state structure costs 0.979574 bits in expectation. Second, the apparent rescue — that transition-level admissibility does determine dynamics — is definitional, since the admissible transition set of a deterministic system is literally the graph of its map. The same triviality recurs one rung up for reaction networks, where the hyperedge set is the network by construction, while the projections below it charge 0.9908 bits to the pairwise view and 8.6978 bits to the species view, with twelve distinct networks exhibited sharing one pairwise projection. What survives is not a claim about the primary arity of anything, but a single principle: every description below a system's event arity is a projection, and projections charge $-\log_2$ of the fibre.

The one genuinely empirical invariant found here is a monotone trade-off. Across all 256 maps on four states, the information lost by the state-level summary decreases strictly and monotonically as the map's own crossing price increases: 0.9796 bits of opacity for bijections, falling through 0.7642, 0.6667 and 0.5000, to exactly 0.0000 for total collapse. Reversibility and state-level legibility trade against each other along a measured curve. Within this family, that restates the program's two-layer principle — that no survivor is both reversible and screened — as a continuum rather than a dichotomy; the general theorem is not supplied here and the result is reported as measured, not proven, beyond the enumerated domain. It was sharpened, not weakened, by correcting the toll from an image-size proxy to the true conditional entropy.

Scope of Proof

Every claim below lives in exactly one row.

Domain	Object	Status
Interface basis	(P, R) generates all roles and both prices	PROVEN
Admission price	Idempotent; not additive under conjunction in general	PROVEN
Crossing price	Exactly additive under composition (chain rule)	PROVEN
Preserve ⇔ zero crossing price	One measurement, two readings	PROVEN
State-admissibility insufficiency	Finite maps on small state sets	PROVEN
Arity separation results	Definitional, not empirical	IDENTIFIED
Monotone trade-off	All 256 maps on 4 states; one family	MEASURED

Domain	Object	Status
Generalization of the trade-off across domains	Untested	OPEN
Physical readings (interfaces in nature)	Structural correspondence only	PULL

1. Where This Paper Starts

The predecessor established the following, all of which this paper assumes. Constraints form a complete distributive lattice on finite prefix-closed families, with meet as intersection and join as union, and no additive valuation is possible on it because $f(A) = f(A \wedge A) = 2f(A)$ forces $f \equiv 0$ for any idempotent composition. Execution forms a monoid whose costs accumulate. The two are linked by inverse image, which is a lattice homomorphism preserving both meet and join with zero violations, while forward image fails meet exactly where the transition is non-injective. Ahead-of-time verification is sound if and only if the admissible set is closed under execution, $T(A) \subseteq A$, which is subject reduction reached as a legality condition rather than a hoped-for property. And the interface was found to be part of the morphism rather than a label attached to it: isomorphic machines produce three distinct composites when joined without transporting the identification, collapsing to one when it is carried.

What that paper could not resolve was why the interface layer refused a single accounting. It measured a saturating cost, showed the cost was idempotent rather than accumulating, and correctly declined to call it a toll — but it did not identify what the second quantity was, or whether both were forced. This paper resolves that by defining the interface operationally and measuring each function it must perform.

2. The Operational Definition

The method here follows a directive that has repeatedly outperformed the alternative in this program: do not define the object by what it is, define it by the irreducible work it performs. Naming the object first generates exceptions; naming the function first unifies the examples. A candidate list of five roles — separate, constrain, translate, preserve, price — was proposed, and each was formalized as an independently computable predicate or quantity so that the list could be tested rather than accepted.

2.1 The basis

An interface I over a finite source S and target T is a pair (P, R) where P partitions S with one distinguished cell labelled inadmissible, and R maps the admitted set into T. Write $\mathrm{Adm} = \{x \in S : P(x) \neq \mathrm{reject}\}$ and, for $y \in T$, write the fibre $R^{-1}(y) = \{x \in \mathrm{Adm} : R(x) = y\}$. Nothing else is supplied. The following are then derived quantities, not additional data:

$$\textbf{constrain(I)} \iff |\mathrm{Adm}| < |S|$$

$$\textbf{translate}(I) \iff \exists\, x \in Adm \text{ with } R(x) \neq x$$

$$\textbf{preserve}(I) \iff |R^{-1}(y)| = 1 \text{ for every } y \text{ in the image}$$

Three interfaces of different character were built from (P, R) alone and their roles read off. The derivation is mechanical: no role required information outside the basis.

Interface (P, R)	constrain	translate	preserve	A bits	X bits
Even-only admission, Gray recoding	True	True	True	1.0000	0.0000
All admitted, quantize 16→4	False	True	False	0.0000	2.0000
Admit x < 4, halve	True	True	False	2.0000	1.0000

2.2 One proposed reduction fails

It was proposed that translate implies separate, on the grounds that translation only makes sense if two representations already exist, which would remove separation from the list as derivable. A direct search refutes this. Formalizing separation as the image of the admitted set differing from the admitted set itself, a counterexample appears immediately: an interface admitting {6, 7} and mapping $6 \mapsto 7$, $7 \mapsto 6$ translates every admitted member while landing in the same space. Translation and separation are independent under this formalization. The reduction from five roles to fewer is real, but it does not come from this direction — it comes from Section 4.

3. The Two Bills

The central claim of the proposed role list was that every crossing has a price. Formalizing price as two separately computable quantities shows this is half right, and the half that fails is informative.

3.1 Definitions

For a source measure μ (uniform throughout this paper unless stated) the two prices are:

$$A(I) = -\log_2 [\, \mu(Adm) / \mu(S) \,] \qquad \textbf{the ADMISSION price}$$

$$X(I) = H(X \mid Y) = \Sigma_Y\, p(y) \cdot H(X \mid Y = y) \qquad \textbf{the CROSSING price}$$

where X is the source variable restricted to Adm and Y = R(X). For X uniform on Adm the crossing price reduces to the fibre form $X(I) = \Sigma_Y (|R^{-1}(y)| / |Adm|) \log_2 |R^{-1}(y)|$, but Section 5 shows that this reduction is exactly what fails when the source is not uniform, and that mistaking the reduction for the definition produces a false composition law.

3.2 The crossing price is not forced

Five interfaces were measured on a sixteen-element source. The third row is the counterexample the role list did not anticipate.

Interface	sep	con	tra	pre	A bits	X bits
Gray recode (bijection)	T	F	T	T	0.0000	0.0000
Quantizer 16 → 4	T	F	T	F	0.0000	2.0000
Validator: even only, injective	T	T	T	T	1.0000	0.0000
Validator + quantizer	T	T	T	F	1.0000	1.0000
Null (identity, no constraint)	F	F	F	T	0.0000	0.0000

Row three separates, constrains — rejecting half the source — translates, and preserves, and its crossing price is exactly zero. A genuine interface with a free crossing therefore exists, and it is not exotic: it is every validating interface whose translation is injective on what it admits. The engineering instance is immediate. A type system rejects ill-typed programs, which is the admission price, and costs nothing at runtime for well-typed ones, which is the zero crossing price. This is the same fact as the predecessor's closure condition, now measured at the boundary instead of inside the machine.

3.3 Preserve and the crossing price are one measurement

Reading the last two columns of the table across every row gives an exact biconditional: preserve holds if and only if the crossing price is zero. These are not two roles that happen to correlate. What survives the crossing and what the crossing costs are the same quantity measured from opposite ends, since preservation is injectivity on the admitted set and injectivity is precisely the vanishing of the conditional entropy. The list of five roles therefore reduces to four by measurement, and the reduction is forced:

$$\textbf{preserve}(I) \iff X(I) = 0$$

4. The Two Algebras, Identified by Their Failure Modes

The predecessor established two algebras and could not say which quantity belonged to which. The two bills answer it, and they are distinguished not by what they satisfy but by what they fail.

4.1 The admission price is idempotent

Applying a validator twice admits exactly what applying it once admits, so the second application is free. Measured across the test interfaces, the admission price of applying I twice equals that of applying it once in every case: 1.0000 and 1.0000, 0.0000 and 0.0000, 2.0000 and 2.0000.

$$A(I \circ I) = A(I) \qquad \textbf{idempotence}$$

And the admission price is not additive under conjunction in general. Combining two constraints gives $-\log_2$ of the measure of an intersection, which equals the sum of the separate prices only when the constraints are independent. Admitting even numbers costs 1.0000 bits, admitting $x < 8$ costs 1.0000 bits, and admitting both costs 2.0000 bits — additive here because the two conditions are independent on a sixteen-element source, and not additive in general. This is the signature of a lattice: the operation is intersection, the natural cost is a measure of it, and the cost accumulates only under independence.

$$\mathbf{A(I_1 \sqcap I_2) \; = \; -\log_2 [\, \mu(Adm_1 \cap Adm_2) / \mu(S) \,] \quad \geq \max(A_1, A_2), \; \leq A_1 + A_2}$$

4.2 The crossing price is exactly additive under composition

For a chain of interfaces the crossing prices telescope. For deterministic maps $f : X \to Y$ and $g : Y \to Z$, the chain rule for conditional entropy gives $H(X|Z) = H(Y|Z) + H(X|Y, Z)$, and because Y determines Z the second term is $H(X|Y)$. Hence:

$$\mathbf{X(g \circ f) \; = \; X(f) + X(g) \qquad \text{chain rule}}$$

with the essential proviso that X(g) is measured with Y carrying the pushforward measure of the source through f, not the uniform measure on Y. Verified over eight random chains from a sixteen-element source through an eight-element intermediate to a four-element target, agreement is exact to nine decimals in all eight cases: for instance 1.272783 + 0.844361 = 2.117144 against a directly measured 2.117144, and 1.647783 + 1.154025 = 2.801808 against 2.801808. This is the monoid signature: composition is the operation, and the cost is a homomorphism into the additive reals.

Property	Admission price A(I)	Crossing price X(I)
Operation	meet (intersection of admissible sets)	composition of realizations
Algebra	distributive lattice	monoid
Under repetition	idempotent: $A(I \circ I) = A(I)$	doubles: $X(f \circ f) = 2X(f)$
Under composition	not defined as a chain	exactly additive
When it vanishes	when nothing is rejected	when the realization is injective
Species (predecessor's terms)	filter	meter

5. A False Shortcut, and Why It Matters

The fibre form of the crossing price, $X = \Sigma_Y (|R^{-1}(y)|/n) \log_2|R^{-1}(y)|$, is correct when the source is uniform, because every element of a fibre then carries equal weight and the conditional entropy within a fibre is exactly the logarithm of its size. It is wrong the moment the source is a pushforward, since the intermediate values then carry unequal weights and the within-fibre entropy is strictly less than the logarithm of the fibre size unless those weights happen to be equal.

This paper's own first attempt at the chain rule used the shortcut for both terms and produced a composition law that failed in every trial — reporting, for example, X(f) = 1.266541 and X(g) = 2.235902 against a composite of 2.235902, which is not a sum but a coincidence caused by computing the same quantity twice under two names. The correction is to use full conditional entropy throughout, after which additivity is exact. The episode is reported because the false formula would otherwise have entered this paper as a theorem, and because it identifies the precise condition under which the convenient form may be used: uniform source, deterministic map, no prior composition.

$$\textbf{valid:}\quad X = \Sigma_Y\,(|R^{-1}(y)|/n)\,\log_2|R^{-1}(y)| \qquad \textbf{only for uniform X}$$

$$\textbf{general:}\quad X = H(X|Y) = \Sigma_Y\, p(y)\cdot H(X\mid Y = y)$$

6. The Identity of the Toll and the Selector

A relation may fail to be a bijection in two independent ways: forward branching, where a state has several legal successors and something must select among them, and backward branching, where several states share a successor and the past cannot be recovered. For a deterministic map these are not two phenomena. Decoding is the inverse relation, whose out-degree at y is by definition the map's in-degree at y, so the crossing price of the map and the selector cost of its inverse are the same number:

$$X(f) \;=\; H(X|Y) \;=\; {}^{E}{}_{Y}[\,\log_2|f^{-1}(y)|\,] \;=\; \text{selector}(f^{-1})$$

Verified to machine precision across five maps on a sixteen-element source: the identity map gives 0.000000 both ways, halving gives 1.000000, quartering 2.000000, reduction mod 3 gives 2.420566, and a random map 0.750000. The quantity must be taken in its intensive form, as an expectation over the target, rather than the extensive sum over fibres; the two differ by normalization and only the intensive one equals the conditional entropy. Independence of the two failure modes is genuine: a bijection has zero of both, a crush loses without needing selection, and a forward-branching relation needs selection without losing. The simultaneous vanishing of both is exactly the bijection.

7. Two Demolitions

7.1 State-level admissibility does not determine dynamics

A natural conjecture holds that a system's behaviour is recoverable from the structure of what it admits. Tested against invariant-set structure it fails outright. Enumerating all twenty-four permutations of four states and grouping by their lattices of invariant subsets yields fifteen distinct structures, with the largest class containing six different dynamics that share the admissible-set structure {Ø, {0,1,2,3}} while differing in behaviour. The projection from transition structure to state structure is therefore lossy, and the loss is measurable at 0.979574 bits in expectation, with the six-member fibre losing exactly $\log_2 6$ = 2.5850 bits.

7.2 The rescue is definitional

The obvious repair is to say that admissibility must be read at arity two — over transitions rather than states — and indeed the transition structure separates all twenty-four dynamics with fibres of size one. This is not a theorem. For a deterministic system the admissible transition set is $\{(x, f(x))\}$, which is literally the graph of f, and the graph of a function determines the function by definition. The result was initially reported in this program as a discovery and is corrected here.

The same triviality recurs one rung up. Reaction networks, where an event $\{A, B\} \Rightarrow C$ is a hyperedge rather than an edge, were tested by projection: four thousand sampled networks give 2,430 distinct pairwise projections and only 15 distinct species-level projections, with expected losses of 0.9908 bits and 8.6978 bits respectively, and a concrete collision of twelve distinct networks sharing one pairwise projection — the pairwise view cannot see which partner a reaction required. But the full hyperedge set is the network by construction, exactly as the graph is the function, so arity three is definitional there for the same reason arity two was definitional for maps.

What survives is not a claim about the primary arity of anything. Each domain has a minimal arity at which its admissible-event set simply is its dynamics, and that arity is fixed by what the domain counts as one legal event — which is an interface property, decided before any execution. Below that arity every description is a projection, and every projection charges $-\log_2$ of the fibre. The ladder is real; the tolls on it are the empirical content; the arity itself is stipulated rather than discovered.

8. The Monotone Trade-off

One genuinely empirical invariant emerged. Taking all 256 maps on four states, grouping them by their own crossing price, and measuring how much information the state-level summary loses about each, the relationship is strictly monotone decreasing.

Crossing price X(f) = H(X|Y)	Fibre partition	Count	Bits lost by the state summary
0.00000	(1,1,1,1)	24	0.9796
0.50000	(2,1,1)	144	0.7642
1.00000	(2,2)	36	0.6667
1.18872	(3,1)	48	0.5000
2.00000	(4)	4	0.0000

The more a dynamics destroys going forward, the less its state-level description loses. Bijections are maximally opaque from the state view, at 0.9796 bits, because they preserve everything and their invariant sets are merely unions of cycles; constant maps are fully transparent, at exactly zero, because collapsing everything into one image pins the map completely. This is the program's two-layer theorem

— no survivor is both reversible and screened — appearing as a continuous curve rather than a dichotomy.

The result was strengthened by correcting an error rather than weakened. The first version of this table keyed on $\log_2(n/|\text{image}|)$, an image-size proxy, which yielded four buckets with the third at 0.5714 bits. Rekeying on the true crossing price H(X|Y) splits that bucket into fibre partitions (2,2) at 0.6667 bits and (3,1) at 0.5000 bits — the same image size, different loss — so the effect is carried by the fibre structure and not by the image size, and the proxy had been averaging two distinct classes into one. Five resolved points replaced four, and monotonicity survived.

9. The Conservation Law and the Generator of P

Sections 3 and 4 established the two bills as algebraically distinct. They are not, however, independent quantities. Together with what reaches the target they conserve the source entropy exactly.

$$\mathbf{A(I) + X(I) + H(Y) = \log_2 N}$$

Verified across eighteen configurations spanning three admission levels and every partition shape at two and three cells, the sum is 3.0000 in every row for a source of eight states, with no exception and no residual. The reading is direct: the source carries $\log_2 N$ bits, admission destroys A of them by refusing entry, realization destroys X more by merging what it admits, and H(Y) is what arrives. The two bills are therefore the two channels through which an interface can destroy information, and their sum is the total destruction:

$$\mathbf{A(I) + X(I) = \log_2 N - H(Y)}$$

Admitted	Cells	Fibre sizes	A	X	H(Y)	Sum
8	2	(4,4)	0.0000	2.0000	1.0000	3.0000
8	2	(1,7)	0.0000	2.4564	0.5436	3.0000
8	3	(3,3,2)	0.0000	1.4387	1.5613	3.0000
6	3	(2,2,2)	0.4150	1.0000	1.5850	3.0000
6	2	(1,5)	0.4150	1.9349	0.6500	3.0000
4	3	(1,1,2)	1.0000	0.5000	1.5000	3.0000

9.1 The partition is optimized, not chosen

The conservation law converts the open question — why this partition rather than another — into an optimization with a determinate answer. Minimizing total destruction A + X is identical to maximizing delivered information H(Y), and for a fixed target capacity the maximizer is unique in shape. Searching all partitions at each capacity:

Source	Capacity	Optimal fibre sizes	H(Y) delivered	X paid
8 states	2 cells	(4, 4)	1.0000	2.0000
8 states	3 cells	(3, 3, 2)	1.5613	1.4387
8 states	4 cells	(2, 2, 2, 2)	2.0000	1.0000

The optimum is the balanced partition at every capacity. Equal cells maximize delivered information; unequal cells waste capacity, since a large fibre destroys more than a small one saves. So P is not an arbitrary act of design. Given a capacity, the cost-minimal partition is forced, and it is the one that equalizes its cells — which is why quantizer design converges on equal-mass bins, and why balanced hashing is not merely convenient but optimal in the exact sense measured here.

One boundary must be stated. This is computed for a uniform source, where equal cell counts and equal probability mass coincide. For a non-uniform source the optimizing criterion is equal mass rather than equal count, and the two diverge; that case is not measured here. The conservation law itself is not restricted in this way — it follows from the chain rule for entropy and holds for any source measure — but the specific claim that the optimum has equal-sized cells is a uniform-source result.

10. The Architecture, Assembled

Read as one construction. Potential enters an admission interface, which partitions the source and charges A(I) once, idempotently, in the lattice. What is admitted is realized across the boundary, charging X(I) per crossing, additively, in the monoid, and charging nothing at all when the realization preserves. The realized state then executes, and observation projects the trajectory, charging again by the same $-\log_2$ rule. The interface is not one layer among these; it is the joint at which the lattice hands off to the monoid, which is why it is the only object in the architecture that carries two prices with two different algebras.

Potential → [Partition: A(I), idempotent] → [Realize: X(I), additive] → Execution → Observation

This also gives the precise sense in which implementation is free. Two implementations honouring the same partition and the same realization up to isomorphism are the same interface, because everything an interface does is a function of (P, R). The implementation is a fibre of the quotient that the interface defines, and fibres are exactly what the crossing price counts. That is the same structure as the recovery of a machine's grammar from behaviour reported earlier in this program, relocated from inside the machine to its boundary.

11. Errors and Corrections

Seven errors were made in the course of this work. Three changed a reported result; the rest were narration written before the corresponding computation returned. All are listed because the corrections are load-bearing and two of them improved the findings.

Error	Consequence and correction
Reported that transition-level admissibility separating all dynamics was a discovery	It is definitional: the admissible transition set of a deterministic system is the graph of its map. Corrected in Section 7.2
Keyed the trade-off table on $\log_2(n/\lvert\text{image}\rvert)$ rather than $H(X\|Y)$	The image-size proxy conflated fibre partitions (2,2) and (3,1). Rekeying gave five resolved points instead of four and strengthened the result
Used the fibre-size shortcut for both terms of the chain rule	Produced a false composition law, computing $H(X\|Z)$ twice under two names. Corrected to full conditional entropy; additivity then exact in 8 of 8
Claimed the two branching costs 'move independently' having exhibited only three of four quadrants	The selector-only quadrant was never produced; both branching constructions had identical degree profiles. Supplied by an independent construction
Built a family asserted to be intersection-closed that was not	$\{0,1,2\} \cap \{0,1,3\} = \{0,1\}$ was absent; the lattice test correctly returned False against a narration claiming FORCED
Asserted a binary partition of operations into execution and constraint	At least four classes exist; the initial test used only meets with fixed sets, the case that preserves everything
Stated that a lattice admits no additive valuation unless graded	False as stated. The correct argument needs only idempotence: $f(A) = f(A \wedge A) = 2f(A) \Rightarrow f(A) = 0$

12. What Remains Open

The monotone trade-off is measured on one family: all 256 maps on four states. Whether it generalizes to larger state sets, to non-deterministic relations, or to domains with different event arities is untested, and the magnitudes are not expected to transfer even if the structure does. The independence of separation from translation is established only under one formalization of separation, and a different formalization might collapse it; the reduction of the role list to a two-primitive basis is therefore secure for constrain, translate and preserve, and provisional for separate. Whether an interface can constrain and realize without separating under any reasonable formalization is open.

The generator of P is now characterized for uniform sources: given a capacity, the cost-minimal partition equalizes its cells. For non-uniform sources the criterion becomes equal probability mass rather than equal count, and that case is unmeasured. Whether a physical field admits a partition generated by the same optimization — the bridge from interface algebra to physics — remains the open problem the conservation law makes precise rather than solves. The admission price has been shown idempotent and sub-additive under conjunction, but its behaviour under a genuine lattice join — admitting either of two constraint sets — has not been measured, and the predecessor's finding that join may require a closure operator suggests the disjunctive case will not be symmetric with the conjunctive one. Whether real verification hardware implements constraint operations with the idempotent signature predicted here remains checkable against instruction-set documentation and undone. And the program's outermost wall stands: no completed self-description, since a map identical to its territory is a fixed point and the axiom executes fixed points — a prohibition this program has previously re-derived from below, in the structural impossibility of a protocol being its own complete contract.

Status Ledger

Claim	Status	Basis (live numbers)
(P, R) generates constrain, translate, preserve, A, X	PROVEN	Three interfaces, all roles derived from the basis alone
Crossing price is not forced: free interfaces exist	PROVEN	Validator row: A = 1.0000, X = 0.0000, all four roles hold
preserve(I) ⇔ X(I) = 0	PROVEN	Exact biconditional across all five test interfaces
A(I ∘ I) = A(I)	PROVEN	1.0000/1.0000, 0.0000/0.0000, 2.0000/2.0000
X(g ∘ f) = X(f) + X(g) with pushforward measure	PROVEN	8 of 8 chains exact to 9 decimals
Fibre shortcut invalid for non-uniform sources	PROVEN	Produced a false law; corrected to conditional entropy
X(f) = selector(f^{-1})	PROVEN	Machine precision, 5 maps; 1.000000, 2.000000, 2.420566, 0.750000
Translate does not imply separate	PROVEN	Counterexample: admit {6,7}, map 6↔7
State-admissibility does not determine dynamics	PROVEN	6 dynamics share one invariant-set lattice; 0.979574 bits lost
Transition-admissibility separation is definitional	IDENTIFIED	The admissible set is the graph of the map
Hyperedge-level separation likewise definitional	IDENTIFIED	4,000 networks; 0.9908 and 8.6978 bits lost below event arity
Monotone trade-off: forward collapse vs state opacity	MEASURED	0.9796 → 0.7642 → 0.6667 → 0.5000 → 0.0000, strictly monotone
A + X + H(Y) = $\log_2 N$	PROVEN	18 configurations, sum = 3.0000 exactly in every row
Balanced partition maximizes delivered information	PROVEN	Optimal at every capacity: (4,4), (3,3,2), (2,2,2,2)

Claim	Status	Basis (live numbers)
Equal-cell optimum for non-uniform sources	OPEN	Criterion becomes equal mass; not measured
Trade-off generalizes beyond this family	OPEN	One family, 4 states, 256 maps
Separate is independent of the other roles	PROVISIONAL	One formalization only
Admission price under lattice join	OPEN	Not measured; closure may be required
Physical instantiation of the two bills	PULL	Structural correspondence only
Derivation ever completed	FORBIDDEN	Complete self-description is a fixed point

---

Appendix: Live Output, Quoted Verbatim

```
INTERFACE ROLES
 interface                    sep  con  tra  pre  FILTER bits  METER bits
 Gray recode (bijection)        T   F   T   T    -0.0000     0.0000
 Quantizer 16->4                T   F   T   F    -0.0000     2.0000
 Validator (even only, injective)  T   T   T   T     1.0000     0.0000
 Validator + quantizer          T   T   T   F     1.0000     1.0000
 Null (identity, no constraint)   F   F   F   T    -0.0000     0.0000

CHAIN RULE (corrected, full conditional entropy)
  X(f)=H(X|Y)  X(g)=H(Y|Z)     sum    H(X|Z)  additive?
    1.272783    0.844361  2.117144  2.117144     True
    1.647783    1.154025  2.801808  2.801808     True
    1.297180    1.284255  2.581436  2.581436     True
 additive in all cases: True

ADMISSION IDEMPOTENCE
 even-only, Gray recode    once 1.0000   twice 1.0000   equal True
 x<4 only, halve           once 2.0000   twice 2.0000   equal True

TOLL = SELECTOR OF THE INVERSE
 x//2    image  8/16  H(X|Y) = 1.000000   selector(f^-1) = 1.000000   match True
 x mod 3  image  3/16  H(X|Y) = 2.420566   selector(f^-1) = 2.420566   match True

MONOTONE TRADE-OFF (keyed on true toll)
  TRUE toll H(X|Y)    fibres    count  mean fibre  mean bits lost
       0.00000 (1, 1, 1, 1)    24      2.583       0.9796
       0.50000  (2, 1, 1)     144      2.167       0.7642
       1.00000    (2, 2)       36      2.000       0.6667
       1.18872    (3, 1)       48      1.750       0.5000
       2.00000     (4,)         4      1.000       0.0000
 strictly monotone decreasing in the TRUE toll? True
```

```
ARITY PROJECTIONS (reaction networks)
  distinct reaction networks sampled : 4000
  distinct BINARY projections     : 2430   expected bits lost 0.9908
  distinct UNARY  projections     :   15   expected bits lost 8.6978
  max networks sharing one binary projection : 12

CONSERVATION  A + X + H(Y) = log2(N) = 3.0000
  admitted  cells     fibre sizes       A       X     H(Y)   A+X+H(Y)
        8     2         (4, 4)  -0.0000  2.0000  1.0000    3.0000
        8     3      (3, 3, 2)  -0.0000  1.4387  1.5613    3.0000
        6     3      (2, 2, 2)   0.4150  1.0000  1.5850    3.0000
        4     3      (1, 1, 2)   1.0000  0.5000  1.5000    3.0000

OPTIMAL PARTITION AT EACH CAPACITY
  N=8, capacity 2: best H(Y)=1.0000 with fibres (4, 4)
  N=8, capacity 3: best H(Y)=1.5613 with fibres (3, 3, 2)
  N=8, capacity 4: best H(Y)=2.0000 with fibres (2, 2, 2, 2)

TRANSLATE DOES NOT IMPLY SEPARATE
  FOUND: admitted [6, 7]  ->  images [6, 7]
  mapping on admitted: [(6, 7), (7, 6)]
```